Software Engineering HSC Practice Examinations - Validation Report

Syllabus Coverage Analysis

System and Data Modeling Tools

- **Data flow diagrams**: Covered in Exam 1 (Q4, Q22), Exam 2 (Q24)
- **Structure charts**: Covered in Exam 1 (Q12, Q25)
- **Data dictionaries**: Covered in Exam 1 (Q8)
- **Class diagrams**: Covered in Exam 1 (Q13, Q24)
- **Storyboards**: Covered in Exam 1 (Q28), Exam 2 (Q22)
- **Decision trees**: Covered in Exam 1 (Q14, Q27), Exam 3 (Q28)

Programming Paradigms

- **Object-oriented paradigm**: Covered in Exam 1 (Q7, Q24)
- **Logic paradigm**: Covered in Exam 1 (Q11)
- **Imperative paradigm**: Covered in Exam 1 (Q20)
- **Functional paradigm**: Covered in Exam 1 (Q11)

Control Structures

- **Sequence**: Covered in Exam 1 (Q21, Q23)
- **Selection**: Covered in Exam 1 (Q1, Q21)
- **Repetition**: Covered in Exam 1 (Q19, Q21)

Relational Databases

- **SQL**: Covered in Exam 1 (Q9), Exam 2 (Q1, Q5, Q10, Q13, Q19, Q21, Q24, Q26, Q27, Q29)
- **Database design**: Covered in Exam 2 (Q24, Q31, Q32)

Web Development (New Content)

- **Front-end frameworks**: Covered in Exam 2 (Q8, Q22)
- **Cross-site scripting**: Covered in Exam 2 (Q6, Q23, Q28)
- **CSS**: Covered in Exam 2 (Q2, Q7, Q12, Q18, Q22, Q26)

Machine Learning (New Content)

- **Machine learning automation through DevOps**: Covered in Exam 3 (Q4, Q25)
- **Regression algorithms**: Covered in Exam 3 (Q2, Q8, Q19, Q21, Q24)
- **Neural Networks**: Covered in Exam 3 (Q3, Q5, Q10, Q13, Q15, Q16, Q17, Q22, Q23, Q29)
- **Training cycle**: Covered in Exam 3 (Q7, Q22, Q23)
- **Execution cycle**: Covered in Exam 3 (Q17, Q22, Q23)

Methods for Testing a System

- Covered in Exam 1 (Q15, Q23), Exam 2 (Q23, Q28), Exam 3 (Q12, Q26)

Character Representation

- Covered in Exam 1 (Q8), Exam 2 (Q21)

Programming with Python

- Covered in Exam 1 (Q10, Q19, Q21, Q23), Exam 2 (Q30), Exam 3 (Q8, Q10, Q21, Q22, Q23, Q27, Q28, Q29, Q30)

Format Compliance

Examination Structure

- **Total marks**: All exams are 80 marks (✓)
- **Duration**: All exams are 2 hours and 30 minutes including 10 minutes reading time (✓)
- **Section I**: Multiple choice (20 marks, ~30 minutes) (✓)
- **Section II**: Short answer questions (40 marks, ~1 hour 10 minutes) (✓)
- **Section III**: Extended response (20 marks, ~40 minutes, choice between two questions) (✓)

Question Types

- **Multiple choice**: All exams include 20 multiple choice questions (✓)
- **Short answer**: All exams include a range of short answer questions with varying mark values (✓)
- **Extended response**: All exams include two options for extended response questions (✓)

Stimulus Material

- All exams include appropriate stimulus material in questions (✓)
- Stimulus materials include code snippets, diagrams, tables, and scenarios (✓)

Mark Allocation

- All questions clearly indicate mark values (✓)
- Mark allocation is appropriate to question complexity and expected response length (✓)
- Total marks for each exam equals 80 (✓)

Conclusion

The three practice examinations collectively provide comprehensive coverage of the Software Engineering syllabus, including both content carried over from the previous SDD syllabus and new content specific to Software Engineering. All exams follow the HSC format with appropriate structure, mark allocation, and stimulus materials.

The examinations are designed to progressively assess student knowledge and skills: - **Exam 1**: Focuses on core programming concepts and system modeling - **Exam 2**: Emphasizes database concepts and web development - **Exam 3**: Concentrates on machine learning and advanced topics

This distribution ensures that students are assessed on all aspects of the syllabus across the three examinations, with appropriate emphasis on both foundational knowledge and newer content areas.